ASHUTOSH SEMWAL

Data Engineer | AI-Augmented Data Pipelines

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PROFESSIONAL SUMMARY

Data Engineer with 1.5+ years of experience building production-grade, automated data pipelines on Databricks, PySpark, SQL, and Azure for large-scale healthcare data (pharmacy, eligibility, and claims). Architected “datajobs,” an incremental ETL framework using Medallion Architecture with declarative configuration, automated data quality gates, and Unity Catalog integration. Built “pybrv,” an LLM-integrated data validation framework that auto-generates business rule test cases from a user-provided data schema — combining GenAI automation with rule-based data engineering. Published author in an IEEE Scopus-indexed conference on deep learning-based predictive modeling.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL
- **Big Data & Cloud:** PySpark, Spark, Databricks, Azure, ADF, ADLS, Azure DevOps, Unity Catalog, Delta Lake
- **Data Engineering:** ETL/ELT Pipelines, Medallion Architecture (Bronze/Silver/Gold), Incremental & CDC Processing, Data Validation & Quality Frameworks, API-Based Ingestion
- **AI/ML & GenAI:** LLM-Integrated Automation (schema-driven test generation), Deep Learning (CNN), NLP, Feature Engineering, Model Evaluation
- **BI & Visualization:** Power BI, Tableau, Excel

WORK EXPERIENCE

Complere Infosystem, Mohali, Punjab **Jan 2025 – Present**

Data Engineer I — Python, SQL, PySpark, Databricks, Databricks Workflows, Azure, ADF, ADLS, Azure DevOps, Power BI

- **datajobs —** Architected and built a production incremental ETL framework on Databricks Workflows using Medallion Architecture (Bronze→Silver→Gold) with declarative JSON configuration, MD5-based row-hashing for incremental merge logic, and PRE/POST data quality gates (CRITICAL/WARNING severity, record-level reject routing) — enabling automated, config-driven, quality-controlled pipeline deployment across pharmacy, eligibility, and claims datasets.
- **pybrv —** Built a rule-based data parity and validation framework integrated with an LLM: given a user-provided data schema, the framework automatically generates business rule test cases — cutting manual test-authoring effort and accelerating validation coverage across pharmacy and eligibility datasets.
- Built pharmacy-specific data workflows orchestrated via Databricks Workflows on ADLS: a daily incremental job filtering pharmacy-specific records from the gold layer, and a monthly incremental job cleaning, transforming, and reconciling pharmacy-provided drug lists against the internal drug master list; monitored pipeline runs through Azure Data Factory (ADF) and managed deployment/versioning through Azure DevOps.
- Integrated Unity Catalog, batch process tracking, and automated audit-trail population into the ETL framework, improving traceability and governance across pipeline runs; deployed workspace structures programmatically using the databricks-sdk.
- Implemented API-based ingestion pipelines, transforming multi-source healthcare data into analytics-ready structures, and built Power BI dashboards to monitor data quality KPIs, mismatch trends, and pharmacy metrics for business stakeholders.

PROJECTS

- **NYC Taxi Analytics Data Engineering Project —** Built an end-to-end NYC Taxi Analytics pipeline on Databricks using PySpark and Delta Lake, implementing Medallion Architecture, data quality checks, star schema modeling, and workflow orchestration.
- **Plant Health Analyzer** — Built and deployed a Streamlit web app for real-time plant disease detection using a VGG16 CNN; users upload a leaf image and receive an instant disease prediction with a confidence score.

EDUCATION

- **Master of Science (Data Science) —** Chandigarh University, Chandigarh, Punjab
- **Bachelor of Technology (Mechanical Engineering) —** H.N.B. Garhwal University, Uttarakhand

CERTIFICATIONS

Google Data Analytics (Coursera) · Data Mining (NPTEL) · IBM Data Science (Coursera) · Generative AI for Data Scientists (Coursera, IBM)

PUBLICATIONS

“Deep Learning and Traditional Algorithms: A Comparative Study on Predicting Second-Hand Car Prices” — 3rd IEEE ICIDeA Conference 2025, IEEE Xplore, Scopus-Indexed